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Math 208 Major Exam II Term 242

Original exam date: 13 April 2025. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

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1. The rank of the matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 1 & 5 & -9 \\ 2 & 5 & 2 \end{bmatrix}$$

is equal to

(a) 1

(b) 3

(c) 2

(d) 0

(e) 4

2. Given that $y = e^{4x}$ is a solution of the differential equation $y''' - 8y' - 32y = 0$. The general solution of the differential equation is

(a) $y(x) = c_1 e^{4x} + e^{-2x}\,(c_2 \cos x + c_3 \sin x)$

(b) $y(x) = c_1 e^{4x} + e^{2x}\,(c_2 \cos 2x + c_3 \sin 2x)$

(c) $y(x) = c_1 e^{4x} + e^{2x}\,(c_2 \cos x + c_3 \sin x)$

(d) $y(x) = c_1 e^{4x} + e^{-2x}\,(c_2 \cos 2x + c_3 \sin 2x)$

(e) $y(x) = c_1 e^{4x} + c_2 e^{-2x} + c_3 x e^{-2x}$

3. An appropriate form of a particular solution y_p for the non-homogeneous differential equation

$$(D-2)^3(D^2+9)y = x^2 e^{2x} + x \sin 3x$$

is given by $y_p(x) =$

(a) $(Ax^2 + Bx^3 + Cx^4)e^{2x} + (Ex + Fx^2)\cos 3x + (Gx + Hx^2)\sin 3x$

(b) $(Ax^2 + Bx^3 + Cx^4)e^{2x} + (Ex + Fx^2)\sin 3x$

(c) $(Ax^2 + Bx^3 + Cx^4)e^{2x} + (Ex + Fx^2)\cos 3x$

(d) $(Ax^3 + Bx^4 + Cx^5)e^{2x} + (Ex + Fx^2)\cos 3x + (Gx + Hx^2)\sin 3x$

(e) $(Ax^3 + Bx^4)e^{2x} + (Cx + Ex^2)\cos 3x + (Fx + Gx^2)\sin 3x$

4. If the solution space of the system

$$\begin{aligned} x_1 + 3x_2 - 4x_3 - 8x_4 + 6x_5 &= 0 \\ x_1 + 2x_3 + x_4 + 3x_5 &= 0 \\ 2x_1 + 7x_2 - 10x_3 - 19x_4 + 13x_5 &= 0 \end{aligned}$$

consists of all linear combinations of the three vectors

$$u = (\alpha, 2, 1, 0, 0), \quad v = (-1, \beta, 0, 1, 0) \quad \text{and} \quad w = (-3, \gamma, 0, 0, 1),$$

then $\alpha + \beta + \gamma =$

(a) 1

(b) -1

(c) 2

(d) 0

(e) -2

5. If the characteristic polynomial of the matrix

$$A = \begin{bmatrix} 5 & -6 & 3 \\ 6 & -7 & 3 \\ 6 & -6 & 2 \end{bmatrix}$$

is $p(\lambda) = -(\lambda + 1)^2(\lambda - 2)$, then a basis for the eigenspace of $\lambda = -1$ is

$$v_1 = \begin{bmatrix} 1 \\ \alpha \\ 0 \end{bmatrix} \quad \text{and} \quad v_2 = \begin{bmatrix} -1 \\ 0 \\ \beta \end{bmatrix},$$

where $\alpha + \beta =$

(a) 3

(b) -2

(c) 2

(d) 0

(e) -3

6. A linear homogeneous constant-coefficient differential equation which has the general solution

$$y(x) = Ae^{3x} + Bxe^{3x} + C\cos 4x + D\sin 4x$$

is

(a) $y^{(4)} - 6y''' + 25y'' - 96y' + 144y = 0$

(b) $y^{(4)} - 6y''' + 5y'' - 9y' + 14y = 0$

(c) $y^{(4)} + 3y''' - 25y'' - 96y' + 144y = 0$

(d) $y''' + 25y'' - 96y' = 0$

(e) $y^{(4)} + 25y'' + 144y = 0$

7. If $W(x)$ is the Wronskian of the functions

$$f(x) = x, \quad g(x) = x\ln x, \quad h(x) = x^2,$$

then $W(x) =$

(a) $2x$

(b) x

(c) 0

(d) $-x$

(e) $-2x$

8. The general solution of the differential equation

$$3y''' - 7y'' - 7y' + 3y = 0$$

is

(a) $y(x) = c_1e^{-x} + c_2e^{\frac{1}{3}x} + c_3e^{3x}$

(b) $y(x) = c_1e^{x} + c_2e^{\frac{1}{3}x} + c_3e^{-3x}$

(c) $y(x) = c_1e^{x} + c_2e^{-\frac{1}{3}x} + c_3e^{3x}$

(d) $y(x) = c_1e^{-x} + c_2e^{-\frac{1}{3}x} + c_3e^{-3x}$

(e) $y(x) = c_1e^{x} + c_2e^{-\frac{1}{3}x} + c_3e^{-3x}$

9. If $y(x)$ is the solution of the initial-value problem

$$y'' - 4y' + 3y = 0, \quad y(0) = 7, \quad y'(0) = 11,$$

then $y(1) =$

(a) $5e + 2e^3$

(b) $2e + 5e^3$

(c) $2e - 5e^3$

(d) $2e^3 - 5e$

(e) $5e - 2e^3$

10. A particular solution of the differential equation $y'' + y = \tan x$ is given by $y_p(x) =$

(a) $-\cos x \ln|\sec x + \tan x|$

(b) $\cos x \ln|\sec x + \tan x|$

(c) $\sin x \ln|\sec x + \tan x|$

(d) $-\sin x \ln|\sec x + \tan x|$

(e) $\ln|\sec x + \tan x|$

11. An eigenvector associated with the eigenvalue $\lambda = 3$ of the matrix

$$A = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix}$$

is $\begin{bmatrix} a \\ 1 \end{bmatrix}$ where $a =$

(a) -1

(b) 2

(c) 1

(d) -2

(e) 0

12. If $y_p = A + Bx + Cx^2$ is a particular solution of the differential equation

$$5y'' + 4y' + 3y = 2 + x^2,$$

then $27A + 9B + 3C =$

(a) 11

(b) 19

(c) 13

(d) 15

(e) 17

13. Consider the subspace S of $\mathbb{R}^3$ defined by $S = \{(x, y, z) \mid y = -z\}$. A basis of S consists of the vector(s)

(a) $v_1 = (0, 1, 0)$, $v_2 = (0, -1, 1)$

(b) $v_1 = (1, 0, 0)$

(c) $v_1 = (1, 0, 0)$, $v_2 = (0, -1, 1)$

(d) $v_1 = (1, 0, 0)$, $v_2 = (0, 1, 1)$

(e) $v_1 = (0, -1, 1)$

14. If the matrix $A = \begin{bmatrix} 5 & -6 \\ 2 & -2 \end{bmatrix}$ is diagonalizable with a diagonalizing matrix P and a diagonal matrix D such that $P^{-1}AP = D$, then

(a) $P = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(b) $P = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$

(c) $P = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$

(d) $P = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

(e) $P = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$

15. The characteristic polynomial of the matrix

$$\begin{bmatrix} 3 & 0 & 0 \\ -4 & 6 & 2 \\ 16 & -15 & -5 \end{bmatrix}$$

is $p(\lambda) =$

(a) $-\lambda^3 - 4\lambda^2 + 3\lambda$

(b) $-\lambda^3 + 4\lambda^2 + 3\lambda$

(c) $-\lambda^3 + 3\lambda^2 - 4$

(d) $\lambda^3 + 4\lambda^2 + 3\lambda$

(e) $-\lambda^3 + 4\lambda^2 - 3\lambda$

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	Echelon form has rows $(1,2,3)$, $(0,3,-12)$, $(0,0,0)$: rank 2.
2	**(d)**	$(r-4)(r^2+4r+8)$: $r=4,\ -2\pm 2i$.
3	**(d)**	$r=2$ triple: times x^3; $r=\pm 3i$ simple: times x, keep both $\cos 3x$ and $\sin 3x$.
4	**(d)**	RREF: $x_1=-2x_3-x_4-3x_5,\ x_2=2x_3+3x_4-x_5$: $\alpha=-2,\beta=3,\gamma=-1$, sum 0.
5	**(a)**	$A+I\to 2x-2y+z=0$: $\alpha=1,\ \beta=2$, sum 3.
6	**(a)**	$(r-3)^2(r^2+16)=r^4-6r^3+25r^2-96r+144$.
7	**(b)**	$W=x(2\ln x)-(2x\ln x-x)=x$.
8	**(a)**	$(r+1)(3r-1)(r-3)=0$: $r=-1,\frac{1}{3},3$.
9	**(a)**	$r=1,3$: $y=5e^x+2e^{3x}$, so $y(1)=5e+2e^3$.
10	**(a)**	Var. of parameters, $W=1$: $y_p=-\cos x(\ln\lvert\sec x+\tan x\rvert-\sin x)-\sin x\cos x$.
11	**(b)**	$A-3I=\begin{bmatrix}1 & -2\\ 1 & -2\end{bmatrix}$: $a-2=0,\ a=2$.
12	**(c)**	$C=\frac{1}{3},\ B=-\frac{8}{9},\ A=\frac{20}{27}$: $20-8+1=13$.
13	**(c)**	$(x,-z,z)=x(1,0,0)+z(0,-1,1)$: basis $\{(1,0,0),(0,-1,1)\}$.
14	**(c)**	$\lambda=2$: $(2,1)$; $\lambda=1$: $(3,2)$; $P=[\,(2,1)\ (3,2)\,]$, $D=\mathrm{diag}(2,1)$.
15	**(e)**	$(3-\lambda)(\lambda^2-\lambda)=-\lambda^3+4\lambda^2-3\lambda$.